7

Understanding the impact: an explanatory framework

In this section, we present our methodology designed to investigate how both *classical* and *topological* data characteristics influence the performance of the GNN-based recommender. Inspired by related research (Adomavicius and Zhang, 2012; Deldjoo *et al.*, 2020; Malitesta *et al.*, 2024), our objective is to construct an explanatory statistical model capable of discerning how particular dataset attributes relate to recommendation quality. To achieve this, we propose a four-step pipeline: (i) creation of a comprehensive pool of recommendation datasets, (ii) computation of their classical and topological features, (iii) training and evaluation of GNN-based recommenders on these datasets, and (iv) fitting a linear explanatory model to link the observed data characteristics and system performance.

7.1 Step 1: Dataset generation through graph sampling

A robust statistical investigation requires a sufficiently large sample set to ensure stable inferences. Although one could consider multiple large-scale recommendation datasets from the literature, such an approach would be prohibitively time-consuming and computationally expensive, since it entails training and evaluating complex GNN-based models on

numerous large graphs.

To address this limitation, we adopt a strategy akin to previous works (Adomavicius and Zhang, 2012; Deldjoo *et al.*, 2020) but with a crucial difference. Instead of relying on traditional sampling procedures from the literature (Chen *et al.*, 2017; Sachdeva *et al.*, 2022; Wu *et al.*, 2021), we employ dedicated *graph sampling* techniques such as node-dropout and edge-dropout (Shu *et al.*, 2022; Wu *et al.*, 2021). By varying dropout rates, we produce multiple smaller recommendation datasets that retain a resemblance to the original sources while exhibiting a broad spectrum of topological variations.

Starting from each bipartite user-item graph, we repeatedly apply the following procedure: select a random dropout rate between 0.7 and 0.9, choose a sampling strategy (node-dropout or edge-dropout), and remove nodes or edges accordingly. Each iteration yields a new dataset sample, capturing various topological patterns. Following prior works (Adomavicius and Zhang, 2012; Deldjoo *et al.*, 2020), we generate 600 samples per original dataset, resulting in a total of 1,800 sampled datasets. The pseudo-code of the algorithms for Step 1 are reported in Algorithms 1 and 2.

7.2 Step 2: Calculation of classical and topological features

After creating the dataset collection, we compute a model-agnostic set of descriptors that capture both *classical* data properties and *topological* graph structure as those described earlier in this monograph. To improve comparability and numerical stability in subsequent modeling, we apply a base-10 logarithmic transformation to highly skewed quantities and standardize all predictors (z-score).

7.3 Step 3: Training and evaluation of the models

In the third step, we adopt a uniform, model-agnostic evaluation protocol for GNN-based recommender. Each dataset is partitioned into training, validation, and test subsets. When timestamps are available, a chronological hold-out approach is preferred to preserve temporal order; otherwise, a random split is used. Models are trained with early stopping

Algorithm 1: Graph sampling

Input: Bipartite user-item graph $\mathcal{G}$, dropout rate μ, graph sampling strategy *sampling*.
Output: Sampled graph $\mathcal{G}_m$.
if *sampling* == *nodeDropout* **then**
  $N = (U + I) * (1 - \mu)$
  $\mathcal{V}_m \leftarrow \text{Uniform}_N(\mathcal{U} \cup \mathcal{I})$
  $\mathbf{A}_m \leftarrow \text{MaskNode}(\mathbf{A}, \mathcal{V}_m)$
else if *sampling* == *edgeDropout* **then**
  $N = E * (1 - \mu)$
  $\mathcal{E}_m \leftarrow \text{Uniform}_N(\mathcal{E}_{u \to i})$
  $\mathbf{A}_m \leftarrow \text{MaskEdge}(\mathbf{A}, \mathcal{E}_m)$
  $\mathcal{V}_m \leftarrow \text{Induce}(\mathbf{A}_m)$
$\mathcal{G}_m \leftarrow \{\mathcal{V}_m, \mathbf{A}_m\}$
Return $\mathcal{G}_m$.

Algorithm 2: Sub-dataset generation.

Input: Bipartite user-item graph $\mathcal{G}$, number of samples M.
Output: M sampled graphs.
$m \leftarrow 1$
$\mathcal{M} = \{\}$
while $m \leq M$ **do**
  $\mu \leftarrow \text{Uniform}([0.7, 0.9])$
  sampling $\leftarrow \text{Uniform}(\{\texttt{nodeDropout}, \texttt{edgeDropout}\})$
  `// We use Algorithm 1 for graph sampling`
  $\mathcal{M} \leftarrow \mathcal{M} \cup \text{GraphSampling}(\mathcal{G}, \mu,$ *sampling* $)$
  $m \leftarrow m + 1$
end
Return $\mathcal{M}$.

based on the best validation score on a ranking-oriented metric (e.g., Recall@K or NDCG@K), under a fixed compute budget (maximum epochs and patience). Hyperparameters are either set via a lightweight validation-driven search or fixed to canonical defaults to ensure comparability across runs. Final effectiveness is reported on the test set

and, when applicable, averaged over multiple random seeds to reduce variance. The resulting scores constitute the dependent variable $\mathbf{y}$ used by the explanatory model in Step 4.

7.4 Step 4: Constructing an explanatory statistical model

Finally, we employ a statistical model to determine how the aforementioned features (independent variables) explain variations in the recommender system's effectiveness (dependent variable). Consistent with recent literature (Adomavicius and Zhang, 2012; Deldjoo *et al.*, 2020), we adopt a linear regression model to facilitate interpretability. This choice allows us to derive straightforward explanations about the directional relationships and significance of each dataset characteristic.

Formally, let $\mathbf{X}_c \in \mathbb{R}^{M \times C}$ be the matrix of feature values for M samples and C characteristics, and let $\mathbf{y}$ be the vector representing the performance outcomes. The linear model is given by:

$$\mathbf{y} = \boldsymbol{\epsilon} + \theta_0 + \boldsymbol{\theta}_c \mathbf{X}_c, \tag{7.1}$$

where θ_0 is the intercept (representing the expected value of $\mathbf{y}$ for mean-centered data) and $\boldsymbol{\theta}_c$ is the vector of coefficients associated with the features. We estimate these parameters via Ordinary Least Squares (OLS):

$$(\theta_0^*, \boldsymbol{\theta}_c^*) = \min_{\theta_0, \boldsymbol{\theta}_c} \frac{1}{2} \|\mathbf{y} - \theta_0 - \boldsymbol{\theta}_c \mathbf{X}_c\|_2^2. \tag{7.2}$$

By maximizing the R^2 coefficient and analyzing the signs and magnitudes of the estimated coefficients, we reveal which characteristics most strongly correlate with improvements or declines in GNN-based recommendation performance. This approach complements existing guidelines on regression analysis (Gareth *et al.*, 2013) and provides interpretable insights into how data properties influence the effectiveness of next-generation recommender systems.

7.5 Impact of characteristics

In this section, we discuss how to interpret the output of the explanatory framework introduced in Step 4. Rather than focusing on a specific

experimental setting, we provide a general reading guide for understanding how *classical* and *topological* dataset characteristics relate to the performance of GNN-based recommender systems.

Given a linear regression model where each characteristic is associated with a coefficient, the *sign* of a coefficient indicates whether increases in that characteristic tend to be associated with improvements or deteriorations in the chosen performance metric. Positive coefficients suggest a direct correspondence (larger values of the characteristic relate to higher performance), whereas negative coefficients indicate an inverse correspondence.

The *magnitude* of a coefficient, once variables have been properly scaled, offers a notion of relative importance: characteristics with larger absolute coefficients are those for which small changes in the underlying data are more strongly reflected in the model's performance. However, these magnitudes should be interpreted in conjunction with measures of model fit and statistical significance, rather than in isolation.

A third element concerns *statistical significance*. Even when a coefficient has a non-negligible magnitude, it should only be interpreted as evidence of a systematic effect if standard significance tests support it. Non-significant coefficients are better viewed as inconclusive: the available data do not provide enough evidence to claim a stable association between the corresponding characteristic and performance.

Finally, characteristics rarely act independently. Correlations among them, as well as possible interaction effects, may cause changes in one characteristic to co-occur with changes in others. For this reason, interpreting the impact of individual coefficients should always be done with the broader correlation structure in mind, and with reference to the conceptual grouping of characteristics into *classical* (space size, shape, sparsity-related factors) and *topological* (degree, clustering, assortativity) properties.

7.5.1 Classical characteristics

From an interpretive standpoint, classical characteristics offer a coarse-grained description of the user-item space. Properties such as the overall size of the interaction space, the balance between users and items, the

density of observed interactions, and the concentration of activity on a subset of users or items help to characterize how much information is available to the model and how it is distributed.

When coefficients associated with density-related measures are positive, they typically indicate that models benefit from richer interaction data, since additional observations provide more evidence for learning robust user and item representations. Conversely, negative coefficients associated with shape-related measures indicate detrimental imbalances between the number of users and items. For example, scenarios with more users than items (or vice versa) may lead to configurations where the model has a limited chance to observe diverse interaction patterns.

Inequality indices (i.e., user- and item-side Gini coefficients) allow us to reason about how skewed the interaction distributions are. A positive coefficient on the item-side index, for instance, suggests that models tend to perform better when there is a clearer separation between popular and less popular items, which aligns with the tendency of many recommenders to focus on highly interacted items. In contrast, weak or unstable effects on the user-side index indicate that strong heterogeneity in user activity is not necessarily beneficial or harmful per se, and may interact with other characteristics such as sparsity or shape.

Overall, classical characteristics describe the *amount* and *allocation* of available information in the interaction matrix. Their coefficients indicate the extent to which simple aggregate statistics of the data are sufficient to account for variations in model performance.

7.5.2 Topological characteristics

Topological characteristics refine this picture by focusing on the structure of the user–item graph beyond aggregate counts. Average degrees, clustering coefficients, and assortativity capture how interactions are organized in terms of local neighborhoods and connectivity patterns, and thus how well the graph supports message passing.

Coefficients associated with user and item average degree indicate how sensitive a model is to the richness of local neighborhoods. Positive values suggest that having more interactions per node provides the model with more informative neighborhoods from which to aggregate

signals, effectively enabling better representation learning. Differences between user- and item-side degrees can reveal whether it is more beneficial to increase the activity of users (e.g., by providing more feedback per user) or to enhance the catalog-side connectivity.

Clustering coefficients describe the tendency of nodes to form tightly connected triadic structures and, in the bipartite setting, the existence of dense local patterns of shared interactions. Positive coefficients on user- or item-side clustering typically signal that models are able to exploit such local redundancy, as it reinforces the collaborative signal by repeatedly exposing similar neighborhoods. When coefficients differ between user and item clustering, this may indicate an asymmetric sensitivity of the model to user- versus item-centric local structures.

Degree assortativity captures whether high-degree nodes tend to connect to other high-degree nodes, or instead to low-degree ones. Positive coefficients on assortativity suggest that the model benefits from graphs where nodes of similar degree are often connected, as this "lookahead" structure enhances the propagation of signals along high-activity regions of the graph. Negative coefficients, in contrast, indicate scenarios where connecting nodes of very different degrees may lead to better information diffusion for the model under consideration.

Taken together, topological characteristics should be read as describing the *shape* and *organization* of the interaction graph. Their coefficients reveal how different GNN architectures leverage specific structural patterns and where their inductive biases align, or conflict, with the topology of the data.

Final remarks

The framework reveals how characteristics relate to model performance through: (i) coefficient signs, which indicate whether a characteristic leads to performance gains/losses; (ii) its magnitude and significance, which quantify the strength and reliability of this association; (iii) the broader correlation among characteristics, which prevents overly simplistic interpretations. Classical characteristics capture the available information and its distribution, while the topological ones describe the user-item graph.

7.5.3 Application to a topology-aware evaluation study

The methodological principles outlined above can be instantiated in concrete evaluation scenarios. A notable example is the topology-aware study of GNN-based recommender systems presented in the work by Malitesta *et al.*, 2024. In that study, the explanatory framework is applied to real-world recommendation datasets and a set of representative GNN architectures. In particular, it considers LightGCN (He *et al.*, 2020), DGCF (Wang *et al.*, 2020), and SVD-GCN (Peng *et al.*, 2022b) as exemplary GNN-based models, evaluated on Yelp2018 (Peng *et al.*, 2022b), Gowalla (He *et al.*, 2020), and Amazon-Book (Wang *et al.*, 2021).

More specifically, the case study revisits the set of classical and topological characteristics introduced in earlier sections and summarized in Table 1.1. Classical properties summarize space size, shape, density, and inequality of interactions (via Gini coefficients on the user and item sides), following prior work on dataset characterization (Adomavicius and Zhang, 2012; Deldjoo *et al.*, 2020). Then, topological properties capture the structural aspects of the user-item bipartite graph, including the average degree (for users and items), average clustering coefficients (user and item projections), and degree assortativity on both sides.

These characteristics are first analyzed through their empirical correlation structure, in order to identify redundant signals and complementary patterns (see Figure 7.1). Moreover, characteristic values and distributions are computed both on the original datasets and on the collection of generated samples (Table 7.1). As reported in the bottom part of the table, the aggregated statistics obtained from all generated samples span a considerably wider range than those computed on the original datasets alone (top part of the table). This observation highlights the increased topological diversity introduced by the sampling procedure and justifies the generation of multiple dataset samples from each original dataset to cover a broader spectrum of topological settings and recommendation scenarios. The explanatory model is then fitted to relate these characteristics to recommendation performance, and the resulting coefficients are examined across datasets and architectures.

This application illustrates how the generic interpretation guidelines

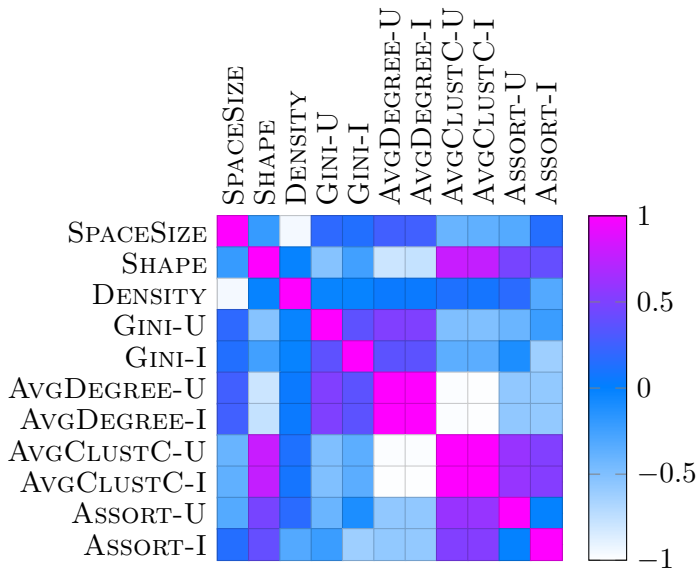


Figure 7.1: Pearson correlation of the selected characteristics and datasets in the study by Malitesta *et al.*, 2024. Many values in $[-0.5, 0.5]$ indicate loose correlation.

proposed in this section are translated into concrete insights. The table of characteristics defines the feature space, the correlation analysis clarifies how these features interact, and the numerical regression results quantify their impact on performance. The reader is referred to that case study for a data-driven instantiation of the framework.

In particular, the results reported in Figure 7.2 offer an illustrative example of this process. When viewed through the lens of our framework, they reveal that degree-related characteristics consistently exert a positive influence on performance, while more complex topological quantities, such as clustering coefficients and assortativity, display model-dependent patterns. At the same time, classical characteristics like density and item-side inequality retain a substantial role, reflecting the interplay between factorization-style effects and neighborhood aggregation in GNN-based recommenders. Rather than focusing on exact numerical values, Figure 7.2 should thus be interpreted as a qualitative map of how different groups of characteristics jointly shape the behavior of topology-aware recommendation models.

Table 7.1: Dataset overall characteristics and aggregated characteristics (minimum and maximum values, mean, and standard deviation) on the sampled sub-datasets. These values refer to the experimental setting adopted in the study proposed by Malitesta *et al.* (2024).

Overall Chars.	**Yelp2018**				**Amazon-Book**				**Gowalla**			
SpaceSize	1.411				1.623				1.544			
Shape	-0.002				0.453				-0.138			
Density	-2.957				-3.318				-3.076			
Gini-U	0.458				0.445				0.467			
Gini-I	0.604				0.603				0.435			
AvgDegree-U	1.434				1.078				1.537			
AvgDegree-I	1.431				1.531				1.399			
AvgClustC-U	-1.561				-1.247				-1.577			
AvgClustC-I	-1.491				-1.606				-1.496			
Assort-U	0.003				-0.044				0.068			
Assort-I	-0.017				0.211				-0.030			
Sample Chars. (600 samples/dataset)	**Min**	**Max**	**Mean**	**Std**	**Min**	**Max**	**Mean**	**Std**	**Min**	**Max**	**Mean**	**Std**
SpaceSize	0.256	1.393	1.000	0.379	0.405	1.593	1.176	0.384	0.430	1.541	1.161	0.375
Shape	0.019	0.105	0.045	0.014	0.325	0.443	0.407	0.021	-0.149	-0.097	-0.129	0.008
Density	-3.699	-2.693	-3.219	0.358	-3.902	-2.896	-3.497	0.365	-3.889	-2.876	-3.380	0.363
Gini-U	0.443	0.508	0.486	0.008	0.384	0.499	0.459	0.023	0.462	0.512	0.491	0.007
Gini-I	0.500	0.609	0.575	0.019	0.518	0.618	0.586	0.018	0.437	0.502	0.478	0.008
AvgDegree-U	0.523	0.926	0.758	0.110	0.318	0.609	0.476	0.077	0.603	1.017	0.846	0.115
AvgDegree-I	0.565	0.955	0.804	0.098	0.682	1.043	0.883	0.096	0.487	0.888	0.717	0.109
AvgClustC-U	-1.144	-0.662	-0.947	0.126	-0.757	-0.407	-0.602	0.095	-1.211	-0.741	-1.013	0.122
AvgClustC-I	-1.092	-0.652	-0.922	0.105	-1.124	-0.751	-0.967	0.099	-1.080	-0.614	-0.881	0.124
Assort-U	-0.051	0.235	0.021	0.035	-0.041	0.533	0.052	0.074	0.042	0.544	0.188	0.071
Assort-I	-0.002	0.237	0.067	0.037	0.000	0.842	0.443	0.264	-0.037	0.161	0.021	0.028

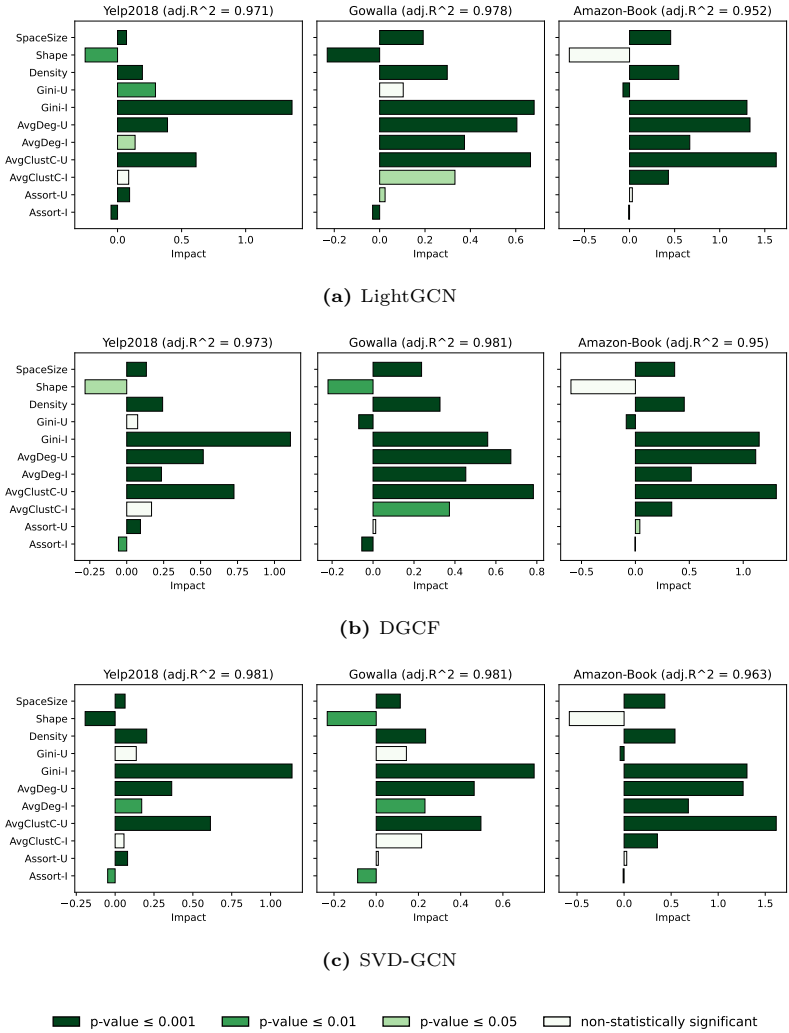


Figure 7.2: Visual representation of the impact of dataset characteristics on the recommendation performance (Recall@20) of GNNs-based recommender systems, for each dataset/model setting in (Malitesta *et al.*, 2024). Bar plot length and direction represent the impact magnitude and whether there is a direct/inverse correspondence between characteristic and performance. Finally, the darker the bar plots, the higher their statistical significance.